

# Evaluating LLM Understanding of Community-Specific Language

**Student:** Guanjun Yan

**Major:** Industrial Engineering

**Faculty Mentor:** Professor Wei Xu

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## 1. Abstract

This project investigated how large language models understand emerging, niche, and coded language in gaming and cybersecurity communities. I reviewed prior work on neologisms, internet slang, semantic change, automated term detection, and domain-specific readability. I then developed a collection and annotation workflow using authentic online community discussions. The project produced a cybersecurity annotation pilot, a Thresh-compatible pipeline, a broader gaming and cybersecurity terminology dataset, and guidelines for labeling linguistic construction, semantic evolution, social function, subcommunity, and comprehension difficulty. My analysis showed that unfamiliar terminology is difficult not only because it is new, but also because its meaning may depend on implicit context, technical knowledge, or insider references. These results motivate future experiments on how context, metadata, retrieval, and self-correction affect LLM understanding.

## 2. Introduction and Project Motivation

Online communities continually create slang, abbreviations, technical expressions, memes, and coded language. These expressions often evolve faster than language-model training cycles and may be difficult for readers or models outside the relevant community.

My project began as a bilingual study of English and Chinese neologisms. Early literature review suggested that Chinese internet language often relies on homophones, character substitution, code-switching, and cultural references, while English expressions frequently use compounding, abbreviations, and semantic shifts. However, I found that word formation alone did not explain comprehension difficulty. A model may recognize every word in a phrase but still miss its community-specific meaning.

I therefore refined the project objective to ask: **What makes gaming and cybersecurity terminology difficult for LLMs to understand, and how can these difficulties be evaluated systematically?**

## 3. Methodology and Research Procedures

I first reviewed research on neologism benchmarks, Chinese internet language, automated term detection, and domain-specific readability. NEO-BENCH [1] provided examples of definition, translation, cloze, and perplexity tasks. NeoN [2] demonstrated how dictionaries, frequency checks, typo detection, and LLM filtering can support automated collection. MedReadMe [3] influenced my approach to measuring comprehension burden in specialized discourse.

I then built a cybersecurity pilot using authentic Reddit discussions. Because Reddit blocked direct collection through Python requests and terminal tools, I retrieved snapshots of Reddit JSON pages, extracted the text, and isolated comment fields. The first pilot contained 193 comment entries, organized into 40 annotation hits across four batches. I tested the labels manually in a spreadsheet before converting the data into a Thresh-compatible format [4].

I later expanded the collection to 12 annotation files averaging approximately 20 hits each. For the broader gaming and cybersecurity dataset, I searched community-specific sources, including gaming forums and cybersecurity discussion boards. I read threads, identified terms or phrases that appeared difficult, niche, or coded, and verified their meanings using surrounding discussion, search tools, Google Translate, and Gemini. I filtered out expressions that were too general or not specific to the target community. For each retained term, I recorded its linguistic construction, subcommunity, authentic usage example, and source.

## 4. Results and Project Outcomes

The project produced four main outcomes:

1. A working pipeline for collecting, cleaning, and preparing online community data.
2. A cybersecurity annotation pilot and a broader gaming and cybersecurity terminology dataset.
3. A Thresh annotation template and supporting repository.
4. Annotation guidelines for linguistic construction, semantic evolution, social function, community, and comprehension difficulty.

One major finding was that novelty and difficulty are not equivalent. A new term may be easy when its meaning is clearly explained, while an older expression may remain difficult because it relies on insider knowledge. I therefore treated comprehension burden separately from the descriptive taxonomy. Easy expressions can be inferred from context, Medium expressions require a targeted lookup, and Hard expressions require broader community research or comparison between possible meanings.

I also found that authentic examples are essential. Some collected sentences directly defined the target term, making them less useful for future evaluation. A stronger benchmark should include natural examples in which the model must infer the meaning rather than copy a definition from the sentence.

## 5. Future Work and Personal Impact

Future work will convert the dataset into controlled evaluation tasks. Models can be tested under progressively richer conditions:

- Term only.
- Sentence context.
- Paragraph context.
- Community metadata.
- External retrieval.

Tasks may include definition generation, meaning selection, paraphrasing, uncertainty reporting, and deciding whether search or clarification is necessary. Future experiments will compare

results across linguistic constructions, semantic changes, subcommunities, and comprehension levels.

A full agentic evaluation could trace failures through uncertainty detection, query formulation, evidence retrieval, reasoning, and self-correction. The project will also prioritize collecting more natural usage examples rather than examples that directly state the definition.

Finally, model errors may occur at several stages. A model may fail to recognize uncertainty, assign an incorrect familiar meaning, search with the wrong query, retrieve irrelevant evidence, or fail to revise its first interpretation. This reframed unfamiliar terminology as both a language-understanding problem and an agentic reasoning problem.

This experience helped me grow from collecting interesting language examples to designing a reproducible research framework. I improved my skills in literature review, dataset construction, annotation design, research engineering, and experimental planning. I also learned that strong research questions often emerge through repeated revision and analysis of real data.

## 6. Acknowledgements

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## 7. References

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